S^{3}ANet: Spatial–Spectral Self-Attention Learning Network for Defending Against Adversarial Attacks in Hyperspectral Image Classification

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***Abstract*— Deep neural networks have demonstrated impressive capabilities in hyperspectral image (HSI) classification tasks. However, they are highly vulnerable to adversarial attacks, raising significant security concerns, especially in the remote sensing community. Even subtle adversarial perturbations that are imperceptible to human observers can mislead deep learning (DL) models and result in incorrect predictions. Therefore, ensuring the robustness of DL models has become a critical focus in addressing security-related remote sensing tasks. Considerable progress has been made in defending against adversarial attacks in HSI classification. Nevertheless, existing methods primarily concentrate on spatial relationships between pixels while overlooking the valuable spectral information present in HSI. Besides, these methods are usually limited to a specific scale and cannot accommodate the precise classification demands for ground objects with various scales. To address these limitations, we propose a spatial–spectral self-attention learning network (S^{3}ANet) for defending against adversarial attacks in HSI classification. Our S^{3}ANet incorporates a pyramid spatial attention learning module to effectively capture spatial dependency at multiple scales. In addition, it utilizes a global spectral transformer to establish correlations between pixels in the spectral dimension. By employing the defense method of spatial–spectral fusion, our model can effectively address adversarial attacks from a comprehensive perspective, seamlessly integrating both spatial and spectral information. Extensive experiments conducted on four benchmark HSI datasets illustrate that the proposed S^{3}ANet achieves competitive performance compared to state-of-the-art methods when faced with adversarial attacks. The code is available online at https://github.com/YichuXu/S3ANet.**

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***Index Terms*— Adversarial attacks and defenses, adversarial example, hyperspectral image (HSI) classification, pyramid spatial attention, transformer.**

I. Introduction

RECENT years have witnessed remarkable progress in hyperspectral imaging systems, enabling more precise Earth observation by harnessing the abundant spectral–spatial information offered by hyperspectral image (HSI) [1], [2], [3]. Consequently, HSI classification, an essential task of assigning semantic labels to image pixels, has gained immensely significant in the remote sensing field [4], [5]. Driven by the success of deep learning (DL) techniques, significant progress has been made in the area of HSI classification [6], [7], [8], [9]. Although DL models have achieved great success in the remote sensing community [10], [11], [12], [13], they usually focus on extracting features from the local receptive field which cause the vulnerability toward adversarial attacks [14]. By incorporating slight adversarial perturbations in the clean image, adversarial examples can be easily created. Despite these adversarial perturbations appearing visually imperceptible from the original images to human observers, they can significantly fool the advanced networks into giving incorrect predictions with high confidence.

The presence of adversarial examples in remote sensing data was first revealed by Czaja et al. [15]. Xu et al. [16] conducted a comprehensive assessment of the susceptibility of DL models toward untargeted and targeted attacks in the context of remote sensing scene classification. They employed adversarial training strategies to defend against the adversarial examples. Two attack methods including the fast gradient sign method (FGSM) and basic iterative method (BIM) are further employed to attack remote sensing image classification models in [17]. Xu and Ghamisi [18] first investigate the universal adversarial examples (UAEs) in remote sensing data. Two feature-level black-box adversarial attacks are further proposed which are more transferable. However, the aim of the UAE proposed in [18] is to deceive the victim models regardless of the predictions. Bai et al. [19] further extended UAE to the targeted attack scenario and proposed two variants of universal adversarial attacks without accessing the victim models in evaluation.

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Recently, Xu et al. [20] discovered that there is a serious threat of adversarial samples in the hyperspectral field, and extended the research on adversarial samples from the RGB field to the hyperspectral domain. Given the limited training data available for HSI classification tasks, deep networks are prone to overfitting and are sensitive to noise. Therefore, it is crucial to design robust deep neural networks that enhance the intrinsic resistibility of DL models against adversarial examples. To this end, they proposed a self-attention context network (SACNet) which captured the global spatial relationships when confronted with adversarial attacks. In this way, the pixels in the whole HSI can build global spatial dependencies and share the perturbation threat. Thus, the SACNet can effectively enhance the inherent resistibility and robustness. Tu et al. [21] further considered the distinctions and commonalities among categories and proposed a robust class context-aware network (RCCA) that utilizes a supervised affinity loss to distinguish intraclass and interclass contextual information. This approach results in more refined global contextual features, thereby enhancing the model's robustness against adversarial samples. Furthermore, Qi et al. [22] employed random input masking and graph theory with self-supervised mechanisms for a robust architecture.

The previous research primarily focuses on building global spatial relationships to enhance the network's inherent resistibility, which is limited to a specific scale and cannot accommodate the precise classification demands for ground objects with various scales. In addition, the networks designed in such a way solely rely on spatial features while ignoring the abundant spectral information contained in HSI. To tackle these challenges, we propose a spatial–spectral self-attention learning network (S^3ANet) to specifically address adversarial attacks in HSI classification. By leveraging the abundant spatial–spectral information contained in HSI, the designed S^3ANet architecture establishes pyramid spatial relationships between pixels and global spectral relationships, achieving adversarial defense from both spatial and spectral perspectives. Since the design of such a spatial–spectral fusion defense network is better aligned with the characteristics of hyperspectral data, it thereby enhances the model's resistibility. The main contributions of this article are summarized as follows.

1) We propose a novel S^3ANet to defend against adversarial attacks in HSI classification. Compared with existing defend networks, our model can simultaneously establish spatial and spectral dependencies between pixels. The experiments on four benchmark HSI classification datasets demonstrate that the proposed S^3ANet can achieve superior performance on the adversarial test set compared with existing state-of-the-art DL models.
2) Given that HSIs may contain different scale objects, thus, establishing spatial relationships at suitable scales becomes crucial. We specially designed a pyramid spatial attention learning module, which captures both global and local spatial context information and simultaneously integrates the dependencies between pixels at different scales.
3) Considering that the transformer structure can effectively build long-distance dependencies, we introduce a global spectral-wise multihead self-attention module which allows the correlations to be established between pixels in the spectral dimension as well.

The article is structured as follows. Section II provides an overview of the related works. Section III presents a detailed description of the proposed S^3ANet. Section IV presents the qualitative and quantitative experimental results along with comprehensive analyses. Finally, Section V concludes the article.

II. Related Works

A. Adversarial Attack

Adversarial attacks typically involve finding adversarial examples against well-trained models. Take the image classification task as an example. Denote a clean image and the corresponding true label as x and y_{true}, respectively. Let $f(x, y_{\text{true}}; \theta)$ be a classifier. The common way to craft adversarial examples for the adversary is to find a perturbation δ, which can maximize the loss function $\mathcal{L}(x+\delta, y_{\text{true}})$ so that $f(x+\delta) \neq y_{\text{true}}$. Accordingly, δ can be obtained by solving the optimization problem as below

$$\delta^* = \arg\max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(x+\delta, y_{\text{true}}; \theta) \tag{1}$$

where $\|\cdot\|_p$ is a regularization term bounded by an L_p norm. In most cases, ϵ is small so that the perturbations are imperceptible to human eyes. To solve (1), gradient-based methods are usually exploited. Three popular methods are introduced below.

1) Fast Gradient Sign Method: In [23], an efficient adversarial attack was proposed by Goodfellow which is termed as FGSM. It generates the adversarial example x_{adv} by maximizing the loss function $\mathcal{L}(x_{\text{adv}}, \hat{y}; \theta)$ with one-step update as follows:

$$x_{\text{adv}} = \text{clip}\big(x - \epsilon \cdot \text{sign}\big(\nabla_x \mathcal{L}\big(x, \hat{y}; \theta\big)\big)\big) \tag{2}$$

where $\hat{y}$ is the target label, $\nabla_x \mathcal{L}(x, \hat{y}; \theta)$ denotes the gradients of the loss function $\mathcal{L}(\cdot)$ with respect to the input image x, clip$(\cdot)$ is the clip function, and sign$(\cdot)$ indicates the sign function.

In [24], the variant of FGSM was proposed by incorporating the ℓ_∞ norm to the generated perturbation

$$x_{\text{adv}} = \text{clip}\left(x - \epsilon \frac{\nabla_x \mathcal{L}\big(x, \hat{y}; \theta\big)}{\big\|\nabla_x \mathcal{L}\big(x, \hat{y}; \theta\big)\big\|_\infty}\right). \tag{3}$$

2) Projected Gradient Descent (PGD): Furthermore, Madry et al. [25] proposed a PGD attack which is a widely used iterative adversarial attack method. PGD attack involves adding a small perturbation in the gradient direction with a step size α at each iteration and adding a random initialization on x. The adversarial example at each iteration can be upgraded as below

$$x_{\text{adv}}^{t+1} = \text{clip}\big(x_{\text{adv}}^t - \alpha \,\text{sign}\big(\nabla_{x_{\text{adv}}} \mathcal{L}\big(x_{\text{adv}}^t, \hat{y}; \theta\big)\big)\big). \tag{4}$$

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3) Carlini and Wagner's Attack (C&W): C&W attacks were proposed by Carlini and Wagner [26] which generate adversarial perturbations by solving a constrained optimization problem with norm restrictions

$$\min \|\delta\|_p + c \cdot f(x+\delta), \quad \text{s.t. } x+\delta \in [0,1]^n \tag{5}$$

where c is a constant value and $f(x+\delta)$ is an objective function, which measures the misclassification probability of the adversarial example.

B. Adversarial Defense

Numerous defense methods have arisen to counter adversarial attacks, with two predominant strategies standing out: adversarial training and modified network architecture.

1) Adversarial Training: An intuitive strategy to perform adversarial defense is adversarial training, which adds adversarial examples in the process of model optimization. Ensemble adversarial training was first introduced by Tramèr et al. [27]. Instead of using a single model, the training data is enhanced with adversarial examples that are generated from several target models. Through comprehensive experiments in the remote sensing scene classification task, Xu et al. [16] demonstrated that adversarial training greatly enhanced the resistance of deep models to adversarial samples. Cheng et al. [28] presented an alternative method for adversarial training that incorporates a generator to model the distributions of adversarial perturbations. A comprehensive review of recent adversarial training methods was given in [29].

2) Modified Network Structure: Adversarial training does not enhance the inherent resistance of the model and also introduces an additional computational burden. To overcome the above shortcomings, recent research has focused on modifying the network structure to enhance its inherent robustness against adversarial attacks. The first study on the effect of network topology on adversarial attacks was done in [30]. They proposed deep contractive networks, which use a smoothness penalty inspired by contractive auto-encoders. This method improves the network's robustness against adversarial examples without sacrificing much performance. In [31], gradient regularization was utilized to enhance the model's robustness toward adversarial attacks. A uniform gradient regularization technique was proposed by Lyu et al. [32], which added a penalty to the loss function for the gradients produced by the adversarial samples. Recently, two robust network architectures were proposed to enhance the robustness against adversarial attacks in HSI classification [20], [21]. However, these methods fail to fully utilize the abundant spectral information contained in HSI, resulting in limited adversarial defense performance.

III. Proposed Method

Most of the current HSI classification defense methods primarily concentrate on establishing global spatial dependencies which are limited to a specific scale and cannot accommodate the precise classification demands for ground objects with various scales. In addition, they usually ignore the abundant spectral information contained in the HSI. To overcome the above problem, a novel S^{3}ANet is proposed in this article. The proposed S^{3}ANet can build relationships from spatial and spectral perspectives simultaneously. We present the proposed S^{3}ANet architecture in Fig. 1.

The proposed S^{3}ANet first utilizes three 3×3 2-D dilated convolutional layers with different dilation rates and one pooling layer. All four layers are composed of $m = 64$ convolutional kernels. Given an HSI data $X \in \mathbb{R}^{h\times w\times c}$ as input, where h, w, and c indicate the height, width, and band numbers, respectively. The shallow features $\boldsymbol{D_3}$ can be calculated as

$$\boldsymbol{D_3} = \mathrm{Dconv}_3\big(P_{\mathrm{avg}}(\mathrm{Dconv}_2(\mathrm{Dconv}_1(X)))\big) \tag{6}$$

where $\mathrm{Dconv}_1(\cdot)$, $\mathrm{Dconv}_2(\cdot)$, and $\mathrm{Dconv}_3(\cdot)$ denote 3×3 2-D convolutional layers with dilation rates 1, 2, and 3, respectively. $P_{\mathrm{avg}}(\cdot)$ represents average pooling with 2×2 kernel size. After that, two proposed modules including pyramid spatial attention learning, and global spectral transformer will be successively introduced.

A. Pyramid Spatial Attention Learning

Inspired by Wang et al. [33] and Huang et al. [34], we design a pyramid spatial attention learning module that builds multiscale spatial relationships. In the pyramid spatial attention learning module, the features D_3 are individually aggregated in each grid using a cross spatial attention module (CSAM). To constrain the scope of the CSAM, we generate grids as virtual boundaries, with each path having a different gridding pooling size. These grids effectively restrict the range of CSAM and facilitate its operation within specific regions. Following PSPNet [35], we set the numbers of grid pooling separately as $(1, 2, 3, 6)$.

For each grid, we apply CSAM to extract spatial dependency at different scales. Take the grid pooling number as 1 for example. As shown in Fig. 2, feature maps $\boldsymbol{Q_1}$, $\boldsymbol{K_1}$, and $\boldsymbol{V_1}$ can be generated through three convolutional 1×1 layers. To obtain the enhanced cross spatial attention, we reshape the features into the following sizes: $\boldsymbol{Q_1} \in \mathbb{R}^{(hw)/4\times m}$, $\boldsymbol{K_1}, \boldsymbol{V_1} \in \mathbb{R}^{((h+w)/2-1)\times m}$. Then, the affinity map $\boldsymbol{A_1}$ can be formulated as

$$\boldsymbol{A_1}(i,j) = \frac{\exp\left(\boldsymbol{Q}_{1_i} \times \boldsymbol{K}^T_{1_j}\right)}{\sum_{k=1}^{\frac{hw}{4}} \exp\left(\boldsymbol{Q}_{1_k} \times \boldsymbol{K}^T_{1_j}\right)} \tag{7}$$

where $\boldsymbol{A_1} \in \mathbb{R}^{(hw)/4\times((h+w)/2-1)}$. We further multiply $\boldsymbol{A_1}$ by $\boldsymbol{V_1}$, and thus we can get the result $\boldsymbol{B_1} = \boldsymbol{A_1} \times \boldsymbol{V_1}$, which is then reshaped to $\boldsymbol{B_1} \in \mathbb{R}^{(h/2)\times(w/2)\times m}$. The output $\boldsymbol{C_1}$ can be obtained by an elementwise addition with the initial features $\boldsymbol{P_1}$. For each path, the CSAM is implemented separately in the corresponding grid to obtain the spatial attention of the pyramid level. We utilize two consecutive CSAM modules to fully aggregate the spatial information. The enhanced attention features denoted as $\boldsymbol{S} \in \mathbb{R}^{(h/2)\times(w/2)\times m}$ are obtained through a nonlinear 1×1 convolutional layer which can be formulated as

$$\boldsymbol{S} = \mathrm{Conv2D}_{1\times1}(\mathrm{Concat}(\boldsymbol{C_1}; \boldsymbol{C_2}; \boldsymbol{C_3}; \boldsymbol{C_4})) \tag{8}$$

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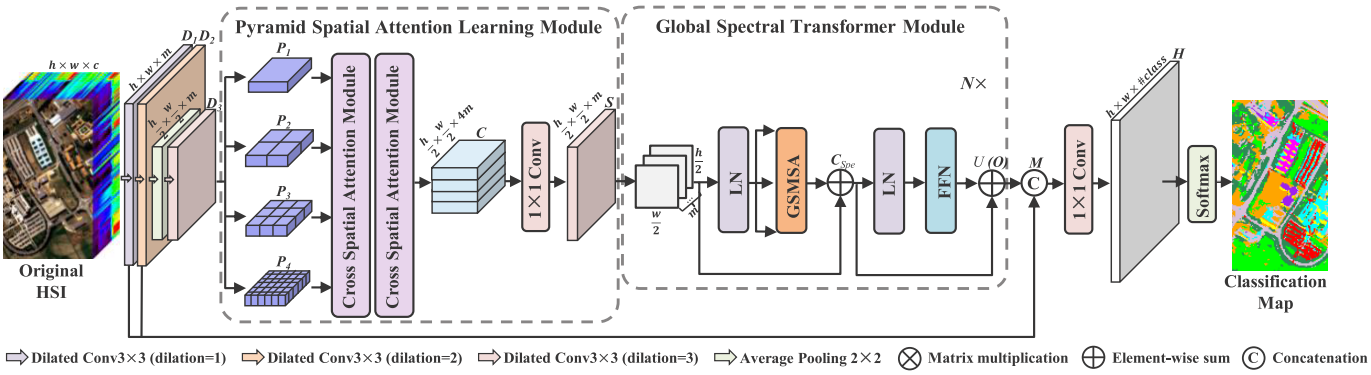

Fig. 1. Architecture of the proposed S^{3}ANet. The shallow features are first extracted through three 3 × 3 2-D dilated convolutional layers with different dilation rates and one pooling layer. Then, we utilize the pyramid spatial attention learning module to build pyramid spatial dependency between pixels at different scales. The global spectral transformer module further captures global spectral dependency. Finally, multiscale classification will be conducted by concatenating the deep features with the first two shallow features.

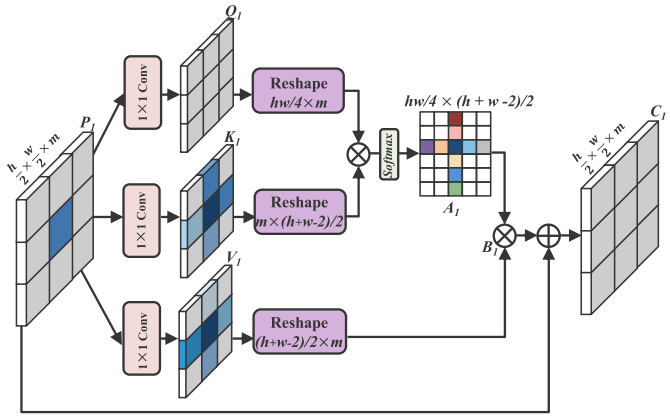

Fig. 2. Illustration of CSAM.

where Conv2D$_{1\times1}$ denotes the nonlinear transformation implemented by an 1 × 1 convolutional layer and $\boldsymbol{C_i}(i = 1, \ldots, 4)$ denotes the results of different paths calculated by CSAM.

B. Global Spectral Transformer

Considering the effectiveness of the transformer in capturing nonlocal long-range dependency and its impressive performance in vision tasks [36], [37], [38], we aim to exploit the transformer to build long-range dependency in spectral dimensions. Since the relationship in spectral dimensions involves the participation of all spectral bands, building the global spectral dependency may also be conducive to enhancing the robustness of the network.

The proposed global spectral transformer mainly includes global spectral multihead self-attention (GSMSA) as shown in Fig. 3, layer normalization (LN), and feed-forward network (FFN). The enhanced spatial attention features $\boldsymbol{S}$ obtained from the pyramid spatial attention learning module are regarded as the input. Our GSMSA first generates query, key, and value by applying a 1 × 1 convolution Conv$_{1\times1}$ followed by a 3 × 3 depth-wise convolution DwConv$_{3\times3}$. The input embedding token $\boldsymbol{Q}_{\mathbf{spe}}$, $\boldsymbol{K}_{\mathbf{spe}}$, $\boldsymbol{V}_{\mathbf{spe}}$ can be obtained

$$\boldsymbol{Q}_{\mathbf{spe}}, \boldsymbol{K}_{\mathbf{spe}}, \boldsymbol{V}_{\mathbf{spe}} = \mathrm{DwConv}_{3\times3}(\mathrm{Conv}_{1\times1}(\boldsymbol{S})) \tag{9}$$

where $\boldsymbol{Q}_{\mathbf{spe}}, \boldsymbol{K}_{\mathbf{spe}}, \boldsymbol{V}_{\mathbf{spe}} \in \mathbb{R}^{n\times m\times(hw)/(4n)}$ and n represents the number of heads in GSMSA. The spectral attention map for each head is further produced

$$\boldsymbol{A}^j_{\mathbf{spe}} = \mathrm{softmax}\left(\boldsymbol{Q}^j_{\mathbf{spe}}\boldsymbol{K}^{j^T}_{\mathbf{spe}}\right) \tag{10}$$

where the function softmax(·) is applied along the last dimension of input. The global spectral attention feature can be obtained by multiplying $\boldsymbol{A}^j_{\mathbf{spe}}$ with $\boldsymbol{V}^j_{\mathbf{spe}}$, and the self-attention of each head can be calculated as

$$\mathrm{head}^j_{\mathbf{spe}} = \boldsymbol{A}^j_{\mathbf{spe}}\boldsymbol{V}^j_{\mathbf{spe}} + \boldsymbol{S}^j. \tag{11}$$

After concatenating the multiple heads, the global spectral features are computed in parallel to capture information from different subspaces in the GSMSA block which can be formulated as

$$\mathrm{GSMSA}\left(\boldsymbol{Q}_{\mathbf{spe}}, \boldsymbol{K}_{\mathbf{spe}}, \boldsymbol{V}_{\mathbf{spe}}\right) = \mathrm{Concat}\left(\mathrm{head}^1_{\mathrm{spe}}, \mathrm{head}^2_{\mathrm{spe}}, \ldots, \mathrm{head}^n_{\mathrm{spe}}\right). \tag{12}$$

Finally, we can get the enhanced spatial–spectral features $\boldsymbol{O} \in \mathbb{R}^{(h/2)\times(w/2)\times m}$ as follows:

$$\boldsymbol{O} = \mathrm{FFN}(\mathrm{LN}(\mathrm{GSMSA}(\boldsymbol{S}))) \tag{13}$$

where FFN(·) and LN(·) present feed-forward network and layer normalization, respectively. Specifically, the FFN is implemented sequentially with a 1 × 1 convolution layer, a GELU activation function, a 3 × 3 depth-wise convolution layer, a GELU activation function, and another 1 × 1 convolution layer.

C. Optimization

After passing through pyramid spatial attention learning and global spectral transformer module, the enhanced feature map $\boldsymbol{O}$ contains both spatial and spectral information. However, directly using $\boldsymbol{O}$ for classification may result in the loss of fine details, particularly at the object boundaries. Therefore, to obtain the fused multiscale feature map $\boldsymbol{M} = [\boldsymbol{D_1}; \boldsymbol{D_2}; U(\boldsymbol{O})] \in \mathbb{R}^{h\times w\times 3\ m}$, the enhanced features $\boldsymbol{O}$ are concatenated with the first two shallow features $\boldsymbol{D_1}$, $\boldsymbol{D_2}$ along with channel dimension. $U(\cdot)$ is an unsampling

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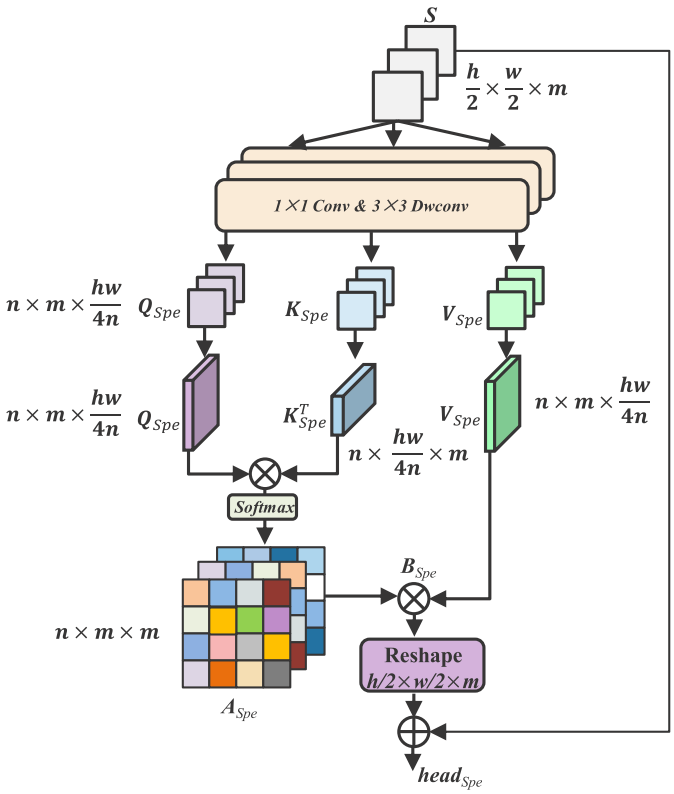

Fig. 3. Illustration of GSMSA.

operation. Finally, the classification map can be calculated as: $\hat{Y} = \text{softmax}(\text{Conv}_{1\times1}(\boldsymbol{M}))$, where $\hat{Y}$ denotes the predicted classification map. The cross-entropy loss is implemented as loss function $\mathcal{L}$

$$\mathcal{L} = -\frac{1}{hw}\sum_{i=1}^{h}\sum_{j=1}^{w}\sum_{c=1}^{C_n} Y_{cij}\log\left(\hat{Y}_{cij}\right) \tag{14}$$

where C_n is the number of classes contained in the HSI and Y indicates the ground truth, respectively. The detailed implementation process performing adversarial attacks on the proposed S^{3}ANet is presented in Algorithm 1.

IV. Experiments

A. Data Description

1) Pavia University: This dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) at Pavia University in Northern Italy in 2001. It has 103 spectral bands. The dataset consists of 610 by 340 pixels and covers nine main categories of interest.

2) Salinas Valley: This dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley in California. It consists of 512 by 217 pixels and has 204 spectral bands. The data contains 16 categories of interest.

3) Houston: This dataset was acquired from the University of Houston campus and its surrounding areas in 2012. It has 144 spectral bands in the wavelength range of 400–1000 nm and a spatial resolution of 2.5 m. It consists of 349 by 1905 pixels and covers 15 categories of interest.

4) Indian Pines: The AVIRIS sensor acquired this dataset in North-Western Indiana in 1992. It has a dimension of 145 by 145 pixels and a spatial resolution of 20 m. It covers 200 spectral bands in the wavelength range of 400–2500 nm after removing water absorption bands. The dataset includes 16 vegetation classes.

Algorithm 1 Adversarial Attack on S^{3}ANet

Input:

1) Benign HSI image X, true label map Y, S^{3}ANet with parameters θ.
2) The perturbation budget $\epsilon = 0.04$, the number of the training epochs $\mathcal{N} = 1000$, the learning rate $\eta = 5e-4$.

1: **for** $iter$ in $range(0, \mathcal{N})$ **do**
2: Calculate the pyramid spatial enhanced features $\boldsymbol{S}$ via (8).
3: Calculate the enhanced global spectral features $\boldsymbol{O}$ via (13).
4: Calculate the cross-entropy loss $\mathcal{L}$ of S^{3}ANet via (14).
5: Update θ by SGD algorithm via $\theta \leftarrow \theta - \eta\nabla_\theta(\mathcal{L})$.
6: **end for**
7: Generate the adversarial image X_{adv} via (3).
8: Input the adversarial image X_{adv} into the optimized S^{3}ANet to obtain the final classification results.

Output: The classification results which predict each pixel on the adversarial set X_{adv}.

B. Experimental Setup

The experiments in this study are all implemented on the PyTorch platform with one NVIDIA GeForce RTX 3090 GPU. To evaluate the effectiveness of defense, we conduct an FGSM attack to generate the adversarial examples via (3). We fix the perturbation budget $\epsilon = 0.04$ and the first class in each dataset as the targeted category. The training set is randomly selected according to the configurations in Tables I–IV. The training process utilizes the Adam optimizer with the initial learning rate of $5e-4$. The training epochs are configured as 1000 on all datasets. The number of heads in GSMSA is fixed at 8. We quantitatively evaluate the classification performance for each method using three commonly adopted evaluation criteria: overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). All experiments were performed with ten random repetitions. The classification results are reported as both the mean value and the standard deviation.

C. Comparisons With the State-of-the-Art Methods

In this section, we present the classification results of the proposed S^{3}ANet along with the nine advanced compare methods. The compared methods including SSFCN [39], CVSSN [40], ViT [41], SpectralFormer [10], SSFTT [42], SSTN [43], and recent well designed defense networks specifically SACNet [20], RCCA [21], and MSSA [22].

Table V reports the quantitative results on the Pavia University dataset. It can be observed that the performance of

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TABLE I
TRAINING AND TESTING SETTINGS OF THE PAVIA UNIVERSITY DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Asphalt		300	6331	6631
2	Meadows		300	18349	18649
3	Gravel		300	1799	2099
4	Trees		300	2764	3064
5	Metal sheets		300	1045	1345
6	Bare soil		300	4729	5029
7	Bitumen		300	1030	1330
8	Bricks		300	3382	3682
9	Shadows		300	647	947
Total			2700	40076	42776

TABLE II
TRAINING AND TESTING SETTINGS OF THE SALINAS VALLEY DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Brocoli green weeds 1		300	1709	2009
2	Brocoli green weeds 2		300	3426	3726
3	Fallow		300	1676	1976
4	Fallow rough plow		300	1094	1394
5	Fallow smooth		300	2378	2678
6	Stubble		300	3659	3959
7	Celery		300	3279	3579
8	Grapes untrained		300	10971	11271
9	Soil vinyard develop		300	5903	6203
10	Corn senesced green weeds		300	2978	3278
11	Lettuce romaine 4wk		300	768	1068
12	Lettuce romaine 5wk		300	1627	1927
13	Lettuce romaine 6wk		300	616	916
14	Lettuce romaine 7wk		300	770	1070
15	Vinyard untrained		300	6968	7268
16	Vinyard vertical trellis		300	1507	1807
Total			4800	49329	54129

TABLE III
CATEGORY AND SAMPLE SETTINGS OF THE HOUSTON DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Grass healthy		300	951	1251
2	Grass stressed		300	954	1254
3	Grass synthetic		300	397	697
4	Trees		300	944	1244
5	Soil		300	942	1242
6	Water		300	25	325
7	Residential		300	968	1268
8	Commercial		300	944	1244
9	Road		300	952	1252
10	Highway		300	927	1227
11	Railway		300	935	1235
12	Parking lot1		300	933	1233
13	Parking lot2		300	169	469
14	Tennis court		300	128	428
15	Running track		300	360	660
Total			4500	10529	15029

TABLE IV
TRAINING AND TESTING SETTINGS OF THE INDIAN PINES DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Alfalfa		30	16	46
2	Corn-notill		50	1378	1428
3	Corn-mintill		50	780	830
4	Corn		50	187	237
5	Grass-pasture		50	433	483
6	Grass-trees		50	680	730
7	Grass-pasture-mowed		20	8	28
8	Hay-windrowed		50	428	478
9	Oats		15	5	20
10	Soybean-notill		50	922	972
11	Soybean-mintill		50	2405	2455
12	Soybean-clean		50	543	593
13	Wheat		50	155	205
14	Woods		30	1235	1265
15	Buildings-Grass-Trees-Drives		50	336	386
16	Stone-Steel-Towers		50	43	93
Total			695	9554	10249

the fully convolutional networks SSFCN, which utilize spatial and spectral information, may not always be outstanding in the adversarial test set. The performance significantly decreases when facing adversarial attacks, with an OA of less than 50%. On the other hand, we compare four methods that utilize advanced transformer structures for HSI classification, namely, ViT, SpectralFormer, SSFTT, and SSTN. Compared to the previous methods that use fully convolutional networks, transformer-based methods demonstrate relatively better results in addressing adversarial attacks. For example, SpectralFormer achieves an OA of more than 80% on three datasets, except for the Indian Pines dataset where it achieves an OA of more than 70%. The phenomenon observed can be attributed to the capability of the transformer architecture to capture global spatial information, which plays a crucial role in mitigating the detrimental impact caused by adversarial samples. In contrast, the traditional fully convolutional networks primarily focus on local information extraction, making them more sensitive to adversarial attacks. SACNet, RCCA, and MSSA, three recent state-of-the-art adversarial defense algorithms, exhibit a relatively limited degree of resilience against adversarial attacks. Owing to the establishment of global dependencies on both spectral and spatial domains, S^{3}ANet achieves the best defense performance by using the proposed pyramid spatial attention learning and global spectral transformer. Similar conclusions can be drawn from Tables VI–VIII on the other three datasets. For all four datasets, S^{3}ANet achieves an OA of more than 93% which is the best result among the compared methods.

The classification maps of different methods on four datasets are presented in Figs. 4–7. While most compared methods

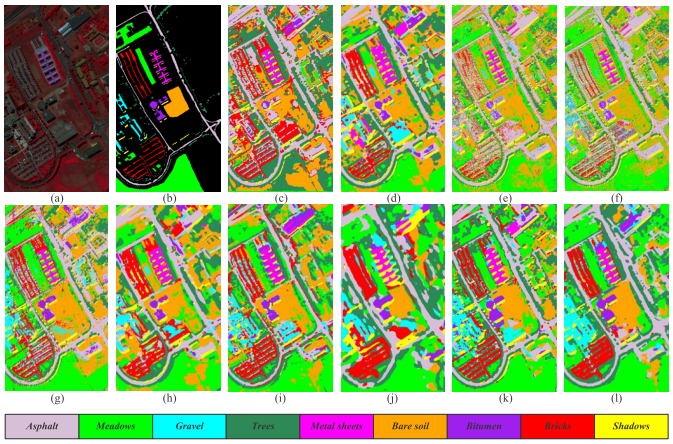

Fig. 4. Classification maps for Pavia University dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) CVSSN. (e) ViT. (f) SpectralFormer. (g) SSFTT. (h) SSTN. (i) SACNet. (j) RCCA. (k) MSSA. (l) Ours.

make significantly wrong predictions under adversarial attacks, the proposed method can obtain satisfactory classification performance which is closest to the ground truth. As illustrated in Fig. 4, we can observe that the middle region of the image belongs to the "Bitumen" class according to the ground truth. However, this region is wrongly classified as either "Asphalt" or "Gravel" by most of the compared methods, which reveals a significant trend of misclassification. In contrast, the proposed S^{3}ANet can achieve accurate segmentation

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TABLE V

QUANTITATIVE CLASSIFICATION RESULTS OF THE PAVIA UNIVERSITY DATASET UNDER THE TARGETED ADVERSARIAL ATTACK. BEST RESULTS ARE SHOWN IN **BOLD**

Class	SSFCN	CVSSN	ViT	SpectralFormer	SSFTT	SSTN	SACNet	RCCA	MSSA	Ours
1	97.68±0.32	98.14±1.32	92.40±2.06	95.11±0.76	95.03±2.99	97.17±1.74	95.99±1.52	96.52±0.32	99.3+0.42	**99.81±0.14**
2	7.39±1.15	98.78±0.64	70.31±5.03	80.83±3.38	94.98±1.26	97.46±1.04	94.11±3.93	95.67±0.79	94.59+7.25	**95.73±1.38**
3	2.03±0.35	80.64±6.91	26.65±6.42	65.99±1.77	86.18±2.83	76.02±11.37	95.81±1.00	94.84±1.47	94.08+6.69	**99.74±0.52**
4	**99.05±0.06**	91.98±6.13	77.88±1.06	90.93±1.37	98.37±0.64	91.36±6.48	97.08±0.85	90.82±0.63	98.58+0.61	97.93±0.19
5	99.89±0.03	96.20±3.44	98.73±0.09	99.13±0.08	99.97±0.09	97.99±1.97	99.88±0.22	97.73±0.80	**100**	99.96±0.05
6	85.23±7.59	70.92±10.70	86.82±3.98	83.66±2.17	96.16±1.32	65.09±10.69	92.25±8.38	97.68±3.64	94.49+10.88	**99.87±0.37**
7	26.94±2.40	82.83±7.76	55.85±2.22	70.07±1.18	96.15±3.43	79.71±11.59	94.19±6.79	98.46±1.60	98.27+0.83	**100**
8	91.31±1.05	90.04±4.44	51.45±9.06	65.51±1.60	87.05±2.76	87.05±4.00	94.02±1.07	**97.74±0.92**	93.93+3.78	96.43±1.24
9	99.77±0.08	93.55±4.21	99.88±0.06	99.17±0.31	99.94±0.08	87.37±7.47	99.69±0.07	96.37±2.58	99.96+0.05	**99.98±0.05**
OA (%)	48.41±1.48	90.80±2.90	73.57±2.50	82.65±1.62	94.54±0.67	87.90±3.09	94.71±2.97	95.98±0.92	95.89+5.03	**97.54±0.72**
Kappa (%)	41.50±1.43	88.13±3.63	65.59±2.95	77.08±1.97	92.65±0.88	84.45±3.84	92.89±3.94	94.59±1.24	94.61+6.55	**96.69±0.96**
AA (%)	67.70±1.23	89.23±2.68	73.33±1.74	83.38±0.59	94.87±0.44	86.58±2.56	95.89±1.88	96.20±0.94	97.02+3.05	**98.83±0.28**
Runtime (s)	**20.70±1.27**	94.36±1.33	921.94±0.46	1125.72±0.66	473.20±0.43	202.23±3.02	49.17±13.12	341.79±0.48	101.81+1.46	277.41±1.22

TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS OF THE SALINAS VALLEY DATASET UNDER THE TARGETED ADVERSARIAL ATTACK. BEST RESULTS ARE SHOWN IN **BOLD**

Class	SSFCN	CVSSN	ViT	SpectralFormer	SSFTT	SSTN	SACNet	RCCA	MSSA	Ours
1	**100**	98.39±2.56	99.94±0.01	99.78±0.17	**100**	99.57±0.76	**100**	**100**	**100**	**100**
2	75.49±4.95	99.97±0.07	71.31±6.35	83.29±6.95	99.97±0.07	99.55±0.80	99.27±0.23	99.98±0.03	99.65+0.51	**100**
3	19.67±11.52	97.48±1.74	92.99±0.98	91.72±4.32	**100**	96.79±1.31	99.71±0.35	**100**	98.52+1.94	**100**
4	99.62±0.09	96.13±2.43	99.09±0.24	99.68±0.24	**99.97±0.04**	94.20±2.17	99.13±0.77	98.73±0.74	99.96+0.05	99.38±0.37
5	17.45±17.56	99.38±0.63	96.05±1.23	86.38±4.46	99.29±0.24	99.36±0.54	**99.52±0.45**	97.68±0.89	94.72+2.02	99.44±0.74
6	99.97±0.02	99.95±0.08	99.72±0.08	99.23±0.14	99.99±0.02	99.93±0.09	99.92±0.04	98.13±0.89	**100**	**100**
7	6.68±0.95	98.45±1.66	93.77±2.40	93.60±2.61	99.92±0.19	98.07±2.43	97.93±1.45	99.14±0.18	**100**	99.99±0.01
8	**99.09±0.54**	91.78±3.22	54.28±2.45	54.21±9.55	84.88±4.91	87.18±8.63	95.59±4.14	96.79±2.57	90.86+5.74	97.83±1.37
9	0.05±0.04	99.54±0.45	93.99±1.90	98.12±0.21	99.99±0.02	99.58±0.28	98.27±1.58	**99.92±0.14**	98.7+1.04	91.98±11.04
10	93.44±2.02	89.60±6.48	76.15±1.17	91.82±0.62	99.15±0.13	92.21±5.81	99.29±0.39	**99.99±0.03**	94.47+2.4	97.54±3.08
11	69.78±2.98	89.46±5.76	92.10±0.89	91.45±2.00	99.96±0.08	91.52±6.33	99.96±0.08	**100**	99.29+0.14	99.99±0.04
12	7.01±1.90	99.78±0.30	64.73±6.80	89.42±6.06	99.90±0.28	**99.83±0.23**	99.99±0.02	99.59±0.11	99.7+0.18	99.17±1.35
13	99.66±0.25	97.03±4.61	98.83±1.68	96.51±3.60	99.72±0.78	98.67±1.84	99.95±0.15	99.92±0.11	**100**	**100**
14	99.97±0.05	96.81±3.74	93.79±0.64	93.57±1.34	99.73±0.53	95.40±5.47	**100**	99.73±0.36	**100**	**100**
15	0.28±0.45	81.10±5.63	80.28±2.73	83.79±3.71	91.31±3.70	81.54±8.27	65.5±38.81	22.74±6.13	84.68+6.63	**99.92±0.12**
16	32.44±6.22	95.44±3.91	98.31±0.15	95.24±1.04	99.80±0.18	95.14±9.07	99.32±1.29	**100**	**100**	**100**
OA (%)	53.13±1.02	93.90±1.06	80.56±0.45	83.44±2.19	95.30±0.71	92.66±2.42	93.63±5.25	88.01±0.64	95.12+2.28	**98.33±1.70**
Kappa (%)	47.00±1.13	93.23±1.17	78.36±0.50	81.58±2.39	94.73±0.79	91.84±2.71	92.79±5.97	86.42±0.73	94.55+2.54	**98.13±1.91**
AA (%)	57.54±1.17	95.64±0.81	87.83±0.84	90.49±1.65	98.35±0.20	95.53±0.96	97.09±2.33	94.52±0.35	97.53+1.00	**99.08±0.97**
Runtime (s)	**12.60±0.99**	163.86±1.69	695.27±5.57	835.73±27.07	79.32±4.46	361.38±8.50	17.34±1.29	92.14±1.87	76.02+1.28	234.13±0.03

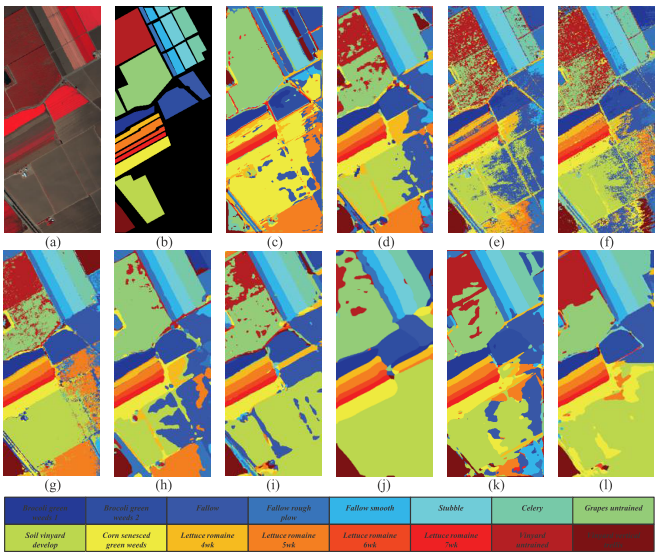

Fig. 5. Classification maps for the Salinas dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) CVSSN. (e) ViT. (f) SpectralFormer. (g) SSFTT. (h) SSTN. (i) SACNet. (j) RCCA. (k) MSSA. (l) Ours.

results where the objects possess continuous smooth surfaces and well-maintained edges, closely resembling the ground truth. This phenomenon demonstrates the robustness of the proposed method against adversarial attacks.

We also report the time cost of each method in Tables V–VIII. The proposed S^{3}ANet has a relatively larger time cost than FCN-based methods, such as SSFCN, because it involves pyramid spatial attention learning and a global spectral transformer that increases the computational complexity of the framework. However, the proposed S^{3}ANet is still competitive with transformer-based methods in terms of time efficiency. For instance, on the Pavia dataset, the proposed S^{3}ANet is faster than ViT and SpectralFormer.

D. Influence of Adversarial Perturbations and Different Attack Methods

In this section, we first perform experiments to compare nine advanced classification methods by varying the ϵ value from 0.01 to 0.1 with a step size of 0.01. In this way, we assess the impact of adversarial attacks with different levels of perturbation on the pixel-level prediction performance. The results on four datasets are presented in Fig. 8. It can be observed that as the value of ϵ increases, the OA values of all methods exhibit a declining trend to different extents. This phenomenon demonstrates the vulnerability of existing classification models to adversarial attacks, especially with larger perturbation intensities. Furthermore, compared to existing defense methods such as SACNet, RCCA, and MSSA, the proposed S^{3}ANet exhibits the strongest resistance to

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TABLE VII
QUANTITATIVE CLASSIFICATION RESULTS OF THE HOUSTON DATASET UNDER THE TARGETED ADVERSARIAL ATTACK. BEST RESULTS ARE SHOWN IN **BOLD**

Class	SSFCN	CVSSN	ViT	SpectralFormer	SSFTT	SSTN	SACNet	RCCA	MSSA	Ours
1	**99.96±0.05**	91.36±8.30	99.56±0.16	95.31±1.43	96.44±1.32	92.69±4.65	91.06±2.31	87.25±2.02	96.19+2.99	95.14±0.28
2	57.71±2.71	91.49±7.04	52.27±1.71	76.08±1.81	**92.47±2.41**	87.75±7.22	83.50±10.64	80.57±6.92	84.47+4.38	86.56±1.80
3	0.00±0.00	95.97±3.60	57.18±5.90	87.10±3.37	99.78±0.23	93.15±8.12	56.62±35.56	99.22±0.40	99.97+0.07	**100**
4	89.38±0.80	96.33±6.34	90.62±0.78	91.66±0.44	94.84±3.35	97.26±2.90	70.76±20.83	89.46±1.48	96.27+1.48	**98.33±0.46**
5	67.76±12.42	96.26±2.19	88.07±1.92	91.79±1.87	98.72±1.60	97.33±2.70	70.59±17.66	96.02±0.61	99.72+0.32	**99.96±0.05**
6	91.20±2.40	92.27±12.47	96.00±1.79	**100**	98.65±0.69	89.69±8.73	79.20±8.16	**100**	**100**	**100**
7	74.13±3.95	89.72±4.29	45.83±3.87	56.56±3.09	79.93±3.83	85.67±3.69	90.83±6.61	83.96±3.51	93.68+1.12	**98.84±0.64**
8	53.49±6.84	**90.40±3.83**	62.42±3.42	75.57±2.00	79.37±4.93	83.85±9.34	66.27±14.23	79.28±4.81	89.36+2.34	85.32±1.65
9	81.50±2.03	87.76±6.58	37.83±2.15	51.91±3.69	63.03±7.82	85.76±4.41	94.31±2.66	94.44±2.46	95.26+2.2	**97.77±0.67**
10	0.00±0.00	**87.73±5.44**	35.86±2.95	67.27±2.37	86.81±4.71	84.68±8.17	57.78±22.68	54.01±15.87	83.11+2.74	79.61±6.79
11	38.67±14.52	88.54±6.37	34.80±4.74	32.42±4.07	79.76±11.30	77.92±9.18	42.70±32.18	48.63±8.58	**97.26+2.28**	93.84±2.22
12	36.91±3.03	87.45±5.31	25.98±1.95	50.93±3.85	73.77±7.63	80.88±9.24	60.09±24.22	91.08±4.81	77.88+3.68	**97.96±0.92**
13	79.23±6.65	92.05±4.54	69.23±2.25	67.57±2.10	88.21±3.64	75.9±13.08	52.90±19.66	94.38±1.86	93.94+2.24	**95.86±0.01**
14	81.17±16.55	90.56±7.64	94.14±0.94	94.14±2.67	99.44±0.38	89.68±6.21	96.88±4.76	**100**	**100**	**100**
15	82.36±3.08	96.36±2.06	93.03±2.17	99.25±0.25	99.79±0.31	93.69±4.26	92.78±6.80	97.81±2.16	**100**	**100**
OA (%)	59.31±1.49	90.74±1.78	59.33±0.42	71.06±0.50	86.63±1.33	87.01±3.35	88.00±1.04	82.36±3.41	92.62+1.38	**93.98±1.07**
Kappa (%)	55.91±1.64	89.99±1.92	56.23±0.46	68.60±0.54	85.54±1.44	85.97±3.62	86.91±1.12	80.78±3.68	92.01+1.50	**93.43±1.17**
AA (%)	62.23±2.39	91.62±1.50	65.52±0.14	75.84±0.59	88.73±1.10	87.73±3.15	87.67±1.11	86.41±2.60	94.07+1.16	**95.28±0.81**
Runtime (s)	60.85±0.75	153.92±0.59	489.09±7.66	589.12±3.85	**23.68±1.24**	338.78±4.49	516.70±6.91	813.11±7.31	161.15+2.71	578.70±1.05

TABLE VIII
QUANTITATIVE CLASSIFICATION RESULTS OF THE INDIAN PINES DATASET UNDER THE TARGETED ADVERSARIAL ATTACK. BEST RESULTS ARE SHOWN IN **BOLD**

Class	SSFCN	CVSSN	ViT	SpectralFormer	SSFTT	SSTN	SACNet	RCCA	MSSA	Ours
1	98.75±2.50	88.27±8.14	96.88±9.38	**100**	98.12±4.00	69.72±17.41	**100**	**100**	**100**	**100**
2	63.15±2.41	82.68±6.59	62.93±1.44	73.94±1.39	83.56±2.09	74.22±8.93	83.08±1.62	82.03±1.40	88.29+1.69	**91.98±0.49**
3	60.23±1.72	85.11±8.09	57.62±3.16	87.06±1.17	82.68±1.66	76.66±11.34	78.64±5.20	89.04±1.93	97.72+0.48	**94.49±0.57**
4	95.12±0.70	88.34±6.40	74.23±1.27	98.69±0.79	84.53±5.34	77.67±8.64	96.04±0.73	98.74±0.69	**100**	96.68±0.21
5	83.02±0.40	92.54±4.90	94.20±1.94	98.23±0.68	97.66±0.66	80.88±8.50	96.47±0.41	89.62±2.00	99.08+0.27	**99.26±0.34**
6	86.16±0.36	98.55±1.04	95.45±0.32	**99.60±0.11**	98.97±0.52	95.94±3.84	95.07±0.98	95.43±0.70	97.47+0.98	97.50±0.50
7	**100**	85.17±16.30	61.25±3.75	**100**	**100**	65.19±18.51	**100**	**100**	**100**	**100**
8	96.70±0.17	99.75±0.61	83.96±2.54	93.14±2.07	99.05±0.48	98.81±1.39	98.76±0.64	**100**	**100**	99.72±0.27
9	**100**	73.36±16.48	90.00±18.44	**100**	96.00±8.00	35.13±9.84	**100**	**100**	**100**	**100**
10	56.41±0.77	81.94±4.97	81.95±2.24	89.88±0.74	81.80±4.30	76.08±5.63	88.23±1.47	90.52±1.85	82.04+13.41	**96.17±0.58**
11	56.94±2.03	**93.91±1.78**	66.10±5.00	66.88±1.28	72.84±2.05	87.73±7.52	76.59±5.15	88.85±2.17	89.5+4.67	93.84±0.80
12	51.76±1.25	74.75±7.32	48.65±3.17	82.37±1.29	85.24±3.28	70.66±9.07	85.45±1.96	87.78±0.87	**92.73+0.5**	91.80±0.64
13	96.51±0.31	98.30±1.49	97.45±0.89	**100**	99.27±0.89	96.40±2.52	99.03±0.43	99.94±0.17	**100**	99.35±0.00
14	77.75±1.35	97.52±1.94	85.05±1.38	94.69±6.68	96.53±0.93	97.46±1.98	92.62±1.71	**98.19±0.79**	96.27+0.88	90.87±0.45
15	82.80±0.64	80.58±7.32	32.62±7.21	43.39±6.67	64.23±7.24	80.81±9.99	98.57±0.99	**100**	**100**	98.99±0.58
16	99.77±0.70	82.05±12.61	99.07±2.13	**100**	99.53±0.93	79.19±8.74	**100**	99.52±0.73	**100**	**100**
OA (%)	68.24±1.35	88.65±1.83	71.10±1.76	80.81±1.36	83.89±1.02	82.36±1.72	86.30±1.26	91.10±0.91	92.63+2.70	**94.49±0.29**
Kappa (%)	64.17±1.51	87.17±2.05	66.80±1.95	78.11±1.53	81.44±1.16	80.06±1.95	84.39±1.41	89.85±1.03	91.54+3.10	**93.68±0.33**
AA (%)	81.57±0.86	87.68±2.92	76.71±2.72	89.24±1.13	90.00±1.00	78.91±1.76	93.03±0.65	94.98±0.50	96.44+1.27	**96.92±0.12**
Runtime (s)	**6.67±0.74**	108.59±3.93	223.36±2.79	145.22±1.86	26.80±1.74	233.87±7.44	10.00±0.72	11.88±1.24	55.83+2.62	195.44±3.38

adversarial examples across different intensities of adversarial attacks. The proposed S^{3}ANet shows the highest and most stable OA under diverse intensities of adversarial attacks on the Pavia University dataset. While most existing methods drop to an OA of less than 60% when ϵ is 0.1, the proposed S^{3}ANet still achieves an OA of over 90%. These results on the four HSI datasets confirm the effectiveness of our proposed method.

We also conducted experiments to evaluate the performance of all methods under different adversarial attack methods beyond FGSM, five typical and efficient attack methods including several novel adversarial techniques commonly employed in remote sensings, such as Mixup [18], Mixcut [18], and TUAE [19]. Adversarial examples were generated by iterating the process ten times using the ℓ_∞ norm with a step size of 0.04 in PGD and C&W attacks. Simultaneously, the classification performance on the clean samples is explored. We present classification accuracy under five attacks and a clean test set in Table IX. We can observe that the ViT method achieves an OA of approximately 73.57% under the FGSM attack on the Pavia University dataset. However, it suffers a dramatic decline to 21.87% and 15.63% when facing PGD and C&W attacks, respectively. Nevertheless, the ViT method maintains an OA of over 64% under three novel adversarial attack methods commonly used in remote sensing. In contrast, the proposed S^{3}ANet consistently achieves an OA of more than 80% across all four datasets except for the Salinas dataset under C&W attack. In addition, we investigate the classification performance of the clean samples. The proposed method achieves the best OA on three datasets, except for the Indian Pines dataset, where it also achieves a relatively high OA of 95.04% which is the second place among all methods. Under different adversarial attacks, our method can still maintain an OA of over 81%, which surpasses the majority of the compared methods. Overall, the proposed S^{3}ANet exhibits better performance than existing state-of-the-art methods on both clean and adversarial test sets, showing strong inherent resistibility.

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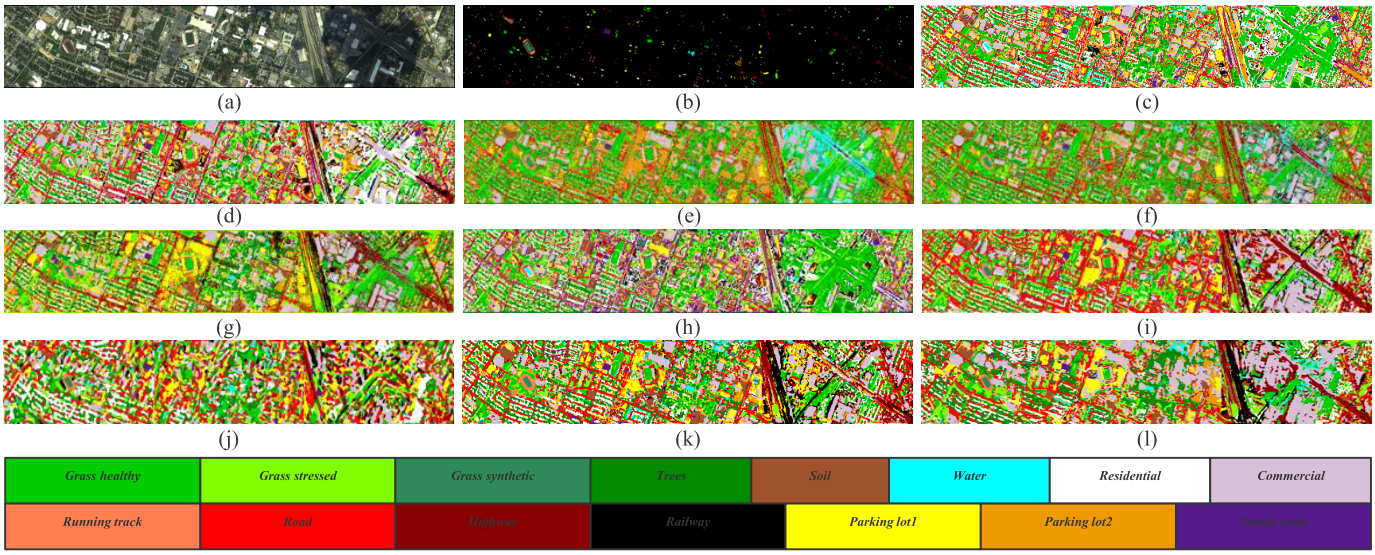

Fig. 6. Classification maps for the Houston dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) CVSSN. (e) ViT. (f) SpectralFormer. (g) SSFTT. (h) SSTN. (i) SACNet. (j) RCCA. (k) MSSA. (l) Ours.

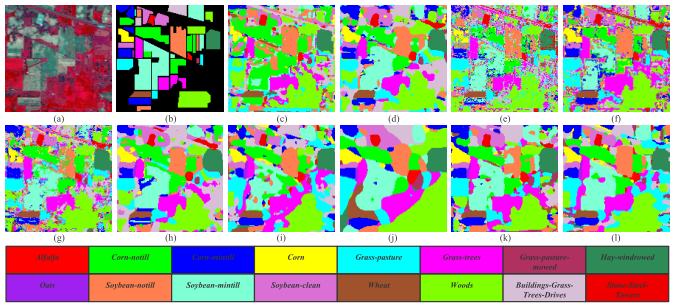

Fig. 7. Classification maps for the Indian Pines dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) CVSSN. (e) ViT. (f) SpectralFormer. (g) SSFTT. (h) SSTN. (i) SACNet. (j) RCCA. (k) MSSA. (l) Ours.

E. Ablation Study and Parameter Analysis

In this section, we evaluate the contribution of each module of the proposed S^{3}ANet to the classification performance. The adversarial examples were created using FGSM with the ℓ_∞ norm, and the perturbation budget ϵ was fixed to 0.04. Table X presents the classification results with different modules. PSA refers to pyramid spatial attention learning, and GST represents the global spectral transformer. As the table clearly indicates, both pyramid spatial attention learning and global spectral transformer can enhance the robustness of the network significantly against adversarial attacks. Pyramid spatial attention learning, moreover, has more benefits in adversarial defense than the global spectral transformer. We observe that the global spectral transformer module enables the network to achieve an OA of 70.76% on the Salinas dataset for instance. While the implementation of pyramid spatial attention learning can elevate the OA to 83.68%. Across all four HSI datasets, the combination of both pyramid spatial attention learning and global spectral transformer yields the highest OAs, indicating the synergistic effect of these two modules in improving classification performance. Overall, the ablation studies demonstrate that both pyramid spatial attention learning and global spectral transformer play crucial roles in enhancing the inherent resistibility of the network. The combination of these modules consistently achieves the best classification results across all datasets.

Fig. 9 shows the average spectral curves of the original HSI data and the adversarial examples produced by different modules of S^{3}ANet. We utilize these results to analyze how each module affects the spectral reflection characteristics of the adversarial samples and the network resistance. By comparing the spectral curves, we can better understand how the clean HSI data and the generated adversarial examples differ, which are easier to perceive by human eyes. This also enables us to intuitively analyze the role of the proposed module in dealing with adversarial threats. As shown in the top row of Fig. 9, the proposed global spectral transformer module significantly narrows the gap between spectral curves. On the other hand, the pyramid spatial attention learning module has a limited impact on reducing such spectral curve differences. However, when both the pyramid spatial attention learning and global spectral transformer modules are combined, the generated adversarial examples exhibit minimal visual differences in spectral curves compared to the original data. This observation holds true for all four datasets, indicating that the combination of pyramid spatial attention learning and global spectral transformer modules achieves the most effective defense performance in terms of minimizing spectral differences.

We further employ the spectral angle [44] to evaluate the distance of the adversarial examples generated by each module from the original HSI. Let x^i and x^i_{adv} be the ith pixel in the original HSI and the adversarial image generated by different modules, respectively. The following formula can be used to compute the spectral angle:

$$\theta\left(x^i_{\text{adv}}, x^i\right) = \cos^{-1}\left(\frac{\left(x^i_{\text{adv}}\right)^T x^i}{\left\|x^i_{\text{adv}}\right\|_2 \left\|x^i\right\|_2}\right). \tag{15}$$

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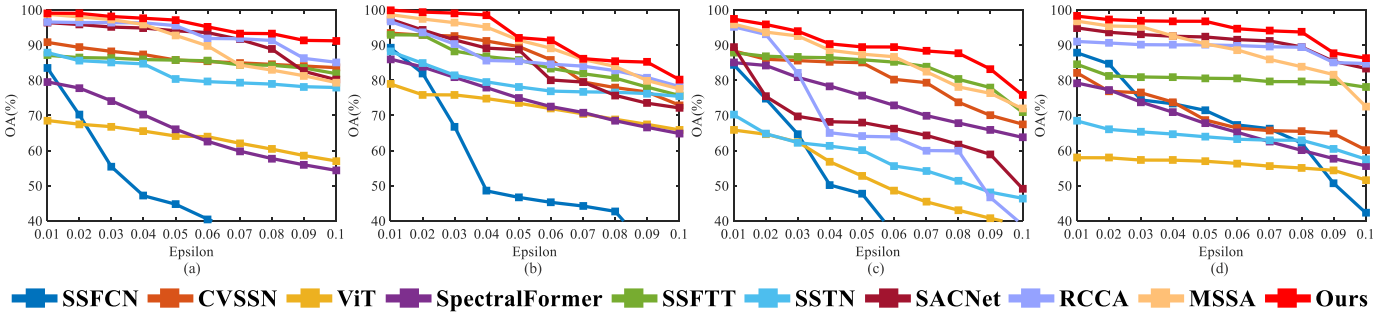

Fig. 8. Classification performance with different adversarial attack perturbation values. (a) Pavia University dataset. (b) Salinas dataset. (c) Houston dataset. (d) Indian Pines dataset.

TABLE IX

CLASSIFICATION ACCURACY OAS (%) UNDER DIFFERENT ATTACK METHODS IN FOUR ORIGINAL CLEAN TEST SETS AND THE ADVERSARIAL TEST SETS. **BOLD** DENOTES THE BEST RESULTS

Dataset	Method	SSFCN	CVSSN	ViT	SpectralFormer	SSFTT	SSTN	SACNet	RCCA	MSSA	Ours
Pavia	Clean Test Set	95.72	97.17	78.89	86.84	98.76	94.77	96.01	96.87	98.79	**99.08**
	FGSM	48.41	90.80	73.57	82.65	94.54	87.90	94.71	95.98	95.89	**97.54**
	PGD	26.49	88.92	21.87	21.13	79.35	78.07	76.92	**85.85**	75.80	83.06
	CW	14.81	84.21	15.63	19.21	61.29	72.69	72.55	77.81	73.15	**80.00**
	Mixup	69.72	82.76	71.53	74.68	77.84	80.44	94.21	92.82	91.37	**96.24**
	Mixcut	68.47	79.48	70.16	73.49	79.66	78.35	87.27	82.30	88.64	**97.79**
	TUAE	44.32	80.52	64.32	77.85	81.07	71.48	93.86	89.76	92.21	**96.85**
Salinas	Clean Test Set	94.47	96.57	88.35	93.10	99.44	94.89	95.21	99.59	99.35	**99.96**
	FGSM	53.13	93.90	80.56	83.44	95.30	92.66	93.63	88.01	95.12	**98.33**
	PGD	18.79	90.62	11.26	12.02	82.59	88.29	81.46	77.50	88.48	**96.24**
	CW	12.78	**81.36**	9.77	11.25	57.64	76.40	74.76	72.72	78.51	65.49
	Mixup	61.77	84.79	80.31	82.65	86.44	82.14	92.94	96.24	96.71	**97.92**
	Mixcut	55.30	88.17	78.46	81.09	84.06	83.02	89.23	90.64	94.33	**95.20**
	TUAE	59.60	92.79	76.07	84.36	85.72	86.43	88.62	88.80	91.89	**96.92**
Houston	Clean Test Set	96.89	97.77	80.67	85.07	97.77	92.43	96.95	98.26	98.64	**99.49**
	FGSM	59.31	90.74	59.33	71.06	86.63	87.01	88.00	82.36	92.62	**93.98**
	PGD	9.59	81.59	7.87	7.34	63.50	74.89	82.66	70.18	78.45	**83.26**
	CW	4.82	80.36	7.26	5.79	47.81	68.27	**83.41**	68.03	80.47	81.04
	Mixup	32.50	78.15	71.73	77.91	81.68	76.84	82.64	81.15	83.56	**84.79**
	Mixcut	41.00	75.82	70.29	72.41	74.36	71.36	79.89	76.32	80.73	**81.45**
	TUAE	42.12	79.23	65.92	75.60	72.46	76.58	74.72	74.54	78.26	**80.03**
Indian	Clean Test Set	75.87	96.85	73.30	83.80	**99.02**	90.08	87.28	93.29	96.37	95.04
	FGSM	68.24	88.65	71.10	80.81	83.89	82.36	86.30	91.10	92.63	**94.49**
	PGD	40.06	83.50	12.08	6.98	76.18	71.05	80.73	81.60	88.21	**89.60**
	CW	31.87	76.18	8.37	6.24	62.85	63.41	82.84	84.52	85.79	**88.01**
	Mixup	68.22	81.42	69.61	70.62	83.17	72.70	84.23	90.07	91.46	**93.25**
	Mixcut	67.96	80.51	66.24	69.07	82.38	71.24	83.27	90.04	89.58	**91.72**
	TUAE	70.43	84.37	51.70	68.89	76.91	78.76	80.14	89.98	90.33	**92.89**

TABLE X

PERFORMANCE CONTRIBUTION OF EACH MODULE IN S^{3}ANET ON FOUR DATASETS (REPORTED IN OA). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. ALL RESULTS ARE IN %

PSA	GST	Pavia University	Salinas	Houston	Indian Pines
–	–	82.86	64.35	49.15	81.60
✔	–	94.80	83.68	84.51	92.10
–	✔	85.17	70.76	83.19	84.46
✔	✔	**97.54**	**98.33**	**93.98**	**94.49**

We then calculate the mean spectral angle with different values of ϵ, highlighting the contribution of each module. The results are exhibited in Table XI, which demonstrate both pyramid spatial attention learning and global spectral transformer effectively improve the defense capabilities of the network in terms of spectral discrepancy.

In the case of the Pavia University dataset, when $\epsilon = 0.1$, the mean spectral angle decreases to 4.11° with the pyramid spatial attention learning module and further decreases to 3.81° with the global spectral transformer module. The combination of pyramid spatial attention learning and global spectral transformer achieves the best performance, with a mean spectral angle of 3.44°. These results align with the disparity observed in the spectral curves shown in Fig. 9. In addition, it can be observed that as the intensity of the adversarial perturbation increases, the mean spectral angle becomes larger. This indicates that the perturbation strength directly impacts the magnitude of the discrepancy.

We visually present the generated adversarial examples and their corresponding adversarial perturbations with different values of ϵ in Fig. 10. For instance, with the Salinas dataset, we can see that the adversarial perturbation is not noticeable

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TABLE XI

PERFORMANCE CONTRIBUTION OF DIFFERENT MODULES IN S^{3}ANET ON FOUR DATASETS USING MEAN SPECTRAL ANGLES WITH DIFFERENT VALUES OF ϵ. **BOLD** INDICATES THE BEST RESULTS

Dataset	PSA	GST	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.1	0.2	0.4	0.6	0.8	1
Pavia University	–	–	0.52	1.04	1.56	2.08	2.59	3.11	3.62	4.12	4.63	5.13	9.97	18.60	25.79	31.67	36.46
	✔	–	0.42	0.83	1.25	1.66	2.07	2.48	2.89	3.30	3.70	4.11	8.09	15.51	22.06	27.64	32.30
	–	✔	0.39	0.77	1.16	1.54	1.92	2.30	2.68	3.06	3.44	3.81	7.50	14.41	20.62	26.14	31.02
	✔	✔	**0.35**	**0.70**	**1.04**	**1.39**	**1.74**	**2.08**	**2.42**	**2.76**	**3.10**	**3.44**	**6.68**	**12.42**	**17.31**	**21.45**	**24.93**
Salinas	–	–	0.47	0.94	1.41	1.88	2.35	2.82	3.29	3.76	4.22	4.69	9.27	17.27	22.78	26.78	29.56
	✔	–	0.45	0.90	1.36	1.81	2.27	2.72	3.18	3.64	4.10	4.56	9.09	17.17	23.73	28.33	31.43
	–	✔	0.36	0.72	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60	7.09	13.53	18.80	22.87	26.24
	✔	✔	**0.25**	**0.49**	**0.74**	**0.99**	**1.23**	**1.48**	**1.73**	**1.97**	**2.22**	**2.47**	**4.89**	**9.51**	**13.76**	**17.56**	**20.87**
Houston	–	–	1.64	3.30	4.97	6.62	8.23	9.77	11.23	12.61	13.90	15.11	24.70	35.74	40.39	42.65	43.99
	✔	–	1.28	2.57	3.86	5.13	6.38	7.60	8.78	9.93	11.03	12.11	21.11	31.98	37.44	40.64	42.73
	–	✔	0.56	1.12	1.68	2.24	2.80	3.34	3.86	4.37	4.87	5.34	9.38	15.25	19.74	23.43	26.50
	✔	✔	**0.47**	**0.93**	**1.40**	**1.86**	**2.31**	**2.75**	**3.19**	**3.61**	**4.03**	**4.43**	**8.04**	**13.58**	**17.74**	**21.06**	**23.82**
Indian Pines	–	–	0.24	0.48	0.73	0.97	1.22	1.46	1.71	1.96	2.21	2.46	5.01	10.27	15.56	20.59	25.14
	✔	–	0.20	0.41	0.61	0.81	1.02	1.23	1.43	1.64	1.85	2.06	4.17	8.55	13.01	17.42	21.65
	–	✔	0.12	0.25	0.38	0.50	0.63	0.75	0.88	1.01	1.13	1.26	2.55	5.17	7.87	10.60	13.36
	✔	✔	**0.11**	**0.22**	**0.33**	**0.44**	**0.55**	**0.66**	**0.77**	**0.89**	**1.00**	**1.11**	**2.21**	**4.42**	**6.59**	**8.71**	**10.76**

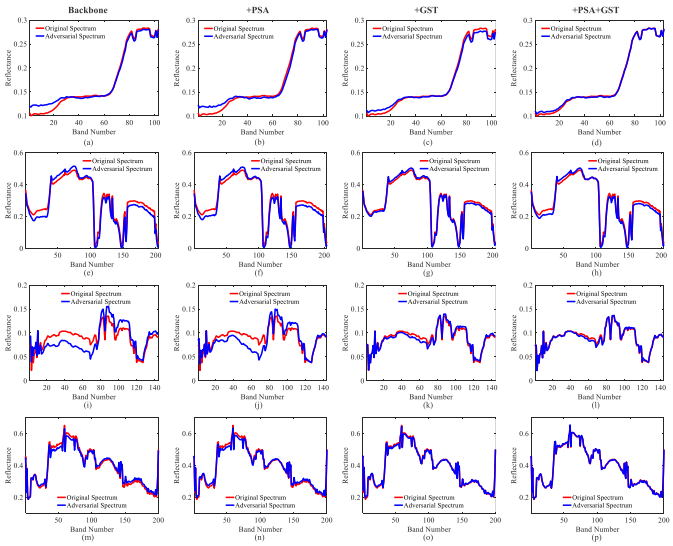

Fig. 9. Average spectral curve of the original HSI and adversarial examples which generated under different modules in S^{3}ANet on four datasets ($\epsilon = 0.1$), from top to bottom: (a)–(d) Pavia University dataset, (e)–(h) Salinas dataset, (i)–(l) Houston dataset, and (m)–(p) Indian Pines dataset.

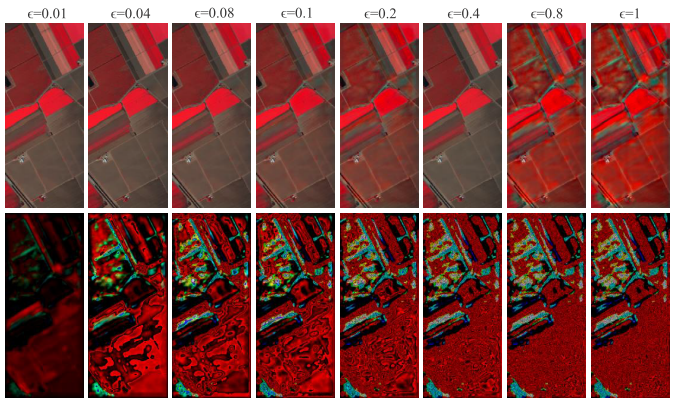

Fig. 10. Visualization of the generated adversarial examples is shown in the first row, while the corresponding adversarial perturbations are displayed in the second row of the Salinas dataset (R: 57, G: 27, and B: 17).

to human observers when ϵ is small (e.g., $\epsilon = 0.01$). However, such subtle differences can greatly fool the current advanced deep neural networks and cause wrong predictions, as Fig. 8 illustrates. This phenomenon emphasizes the severe risk that adversarial attacks bring to the HSI classification task. Therefore, it is crucial to design networks with stronger resistance and greater robustness in HSI classification.

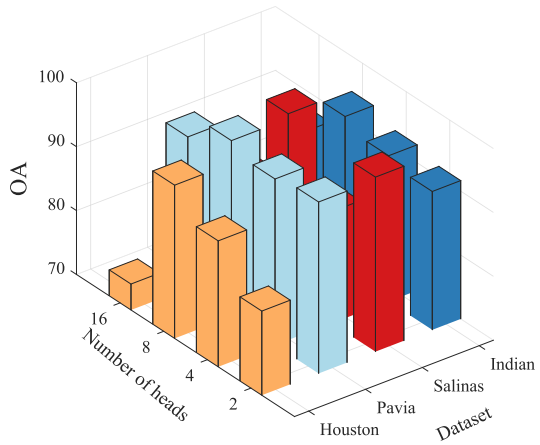

Fig. 11. Classification accuracy of the proposed S^{3}ANet with different number of heads in GSMSA on the four datasets.

The number of heads n in the transformer is one important parameter that influences the performance. We configure n to 2, 4, 8, and 16 to conduct the parameter analysis experiments on four datasets, separately. Fig. 11 presents the OA with a different number of heads n on the four datasets. We can observe that OA values rise as the number of heads n increases. When $n = 8$ the proposed S^{3}ANet achieves the best OAs in all four datasets. However, when n reaches 16, the OA of S^{3}ANet tends to decline. This phenomenon may be attributable to the excessive number of heads, where some attention heads may perform unnecessary or unrelated calculations. Therefore, we choose $n = 8$ for S^{3}ANet in the experiments.

V. CONCLUSION

The vulnerability of deep neural networks toward adversarial examples cannot be neglected since they can fool the advanced deep neural networks to yield incorrect predictions, which has raised serious security issues in the HSI community. While the current research for adversarial defense mainly concentrates on using global spatial relationships, it overlooks the rich spectral information and various object

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scales contained in the HSI. To address these shortcomings, we introduce a novel S^{3}ANet. Our proposed S^{3}ANet first incorporates a pyramid spatial attention learning module to effectively capture spatial dependency at various scales. Furthermore, it utilizes a global spectral transformer to establish relationships between pixels in the spectral dimension. S^{3}ANet can effectively address adversarial attacks from a comprehensive perspective, integrating both pyramid spatial and spectral information. Extensive experiments on four benchmark HSI datasets show that the proposed S^{3}ANet can yield competitive performance compared to state-of-the-art DL methods. In addition, qualitative and quantitative ablation studies from the perspective of spectral characteristics are carried out. The results indicate that the proposed spatial–spectral defense strategy can better handle the risks caused by adversarial attacks.

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